

Module -1

DC Circuit Analysis and Theorems

Dr. Sathya P
Associate Professor
SENSE

Super Node Analysis

Case 1: If a voltage source is connected between the reference node and a non reference node, the voltage at the non-reference node equal to the voltage of the voltage source.

Case 2: If the voltage source (dependant or independent) is connected between two non-reference nodes, the two non-reference nodes form a supernode. Apply both KCL and KVL to determine the node voltages.

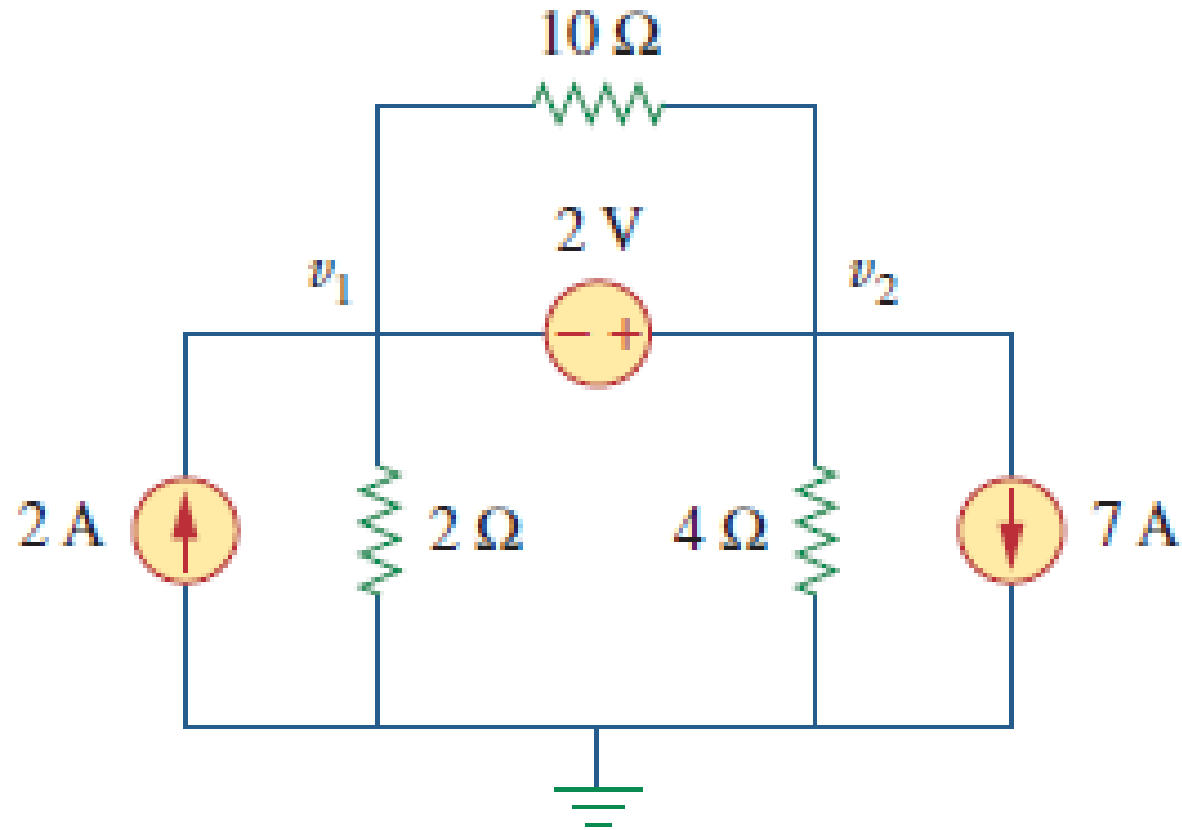
Super node

A **supernode** is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.

- Apply KCL to the supernode by treating both nodes a single supernode while removing any element in parallel to it.
- Apply KVL to the supernode while removing all elements connected to it.

Tutorial - 1

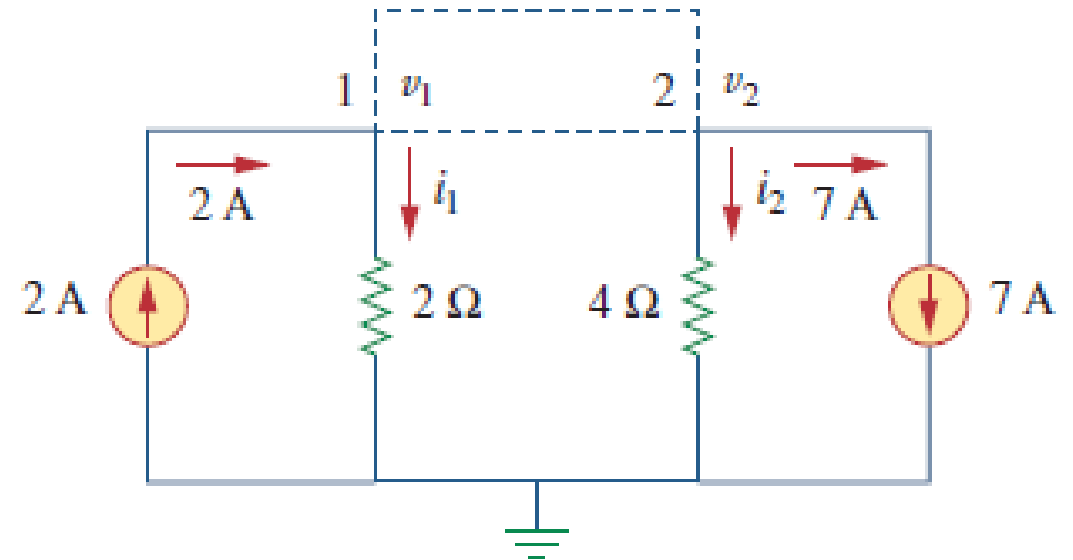
Calculate the node voltages in the circuit.



- Solution:

The supernode contains the 2-V source, nodes 1 and 2, and the 10- Ω resistor. Applying KCL to the supernode

$$2 = i_1 + i_2 + 7$$



Expressing i_1 and i_2 in terms of the node voltages

$$2 = \frac{v_1 - 0}{2} + \frac{v_2 - 0}{4} + 7 \quad \Rightarrow \quad 8 = 2v_1 + v_2 + 28$$

$$v_2 = -20 - 2v_1$$

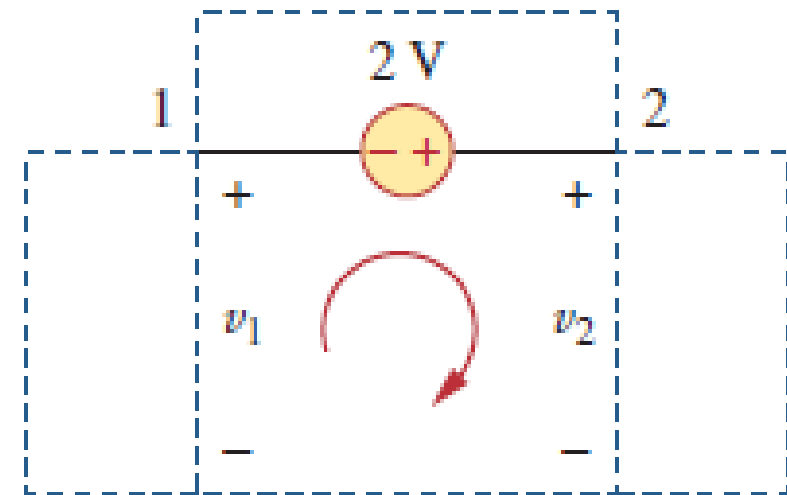
To get the relationship between v_1 and v_2 , we apply KVL to the circuit

$$-v_1 - 2 + v_2 = 0 \quad \Rightarrow \quad v_2 = v_1 + 2$$

$$v_2 = v_1 + 2 = -20 - 2v_1$$

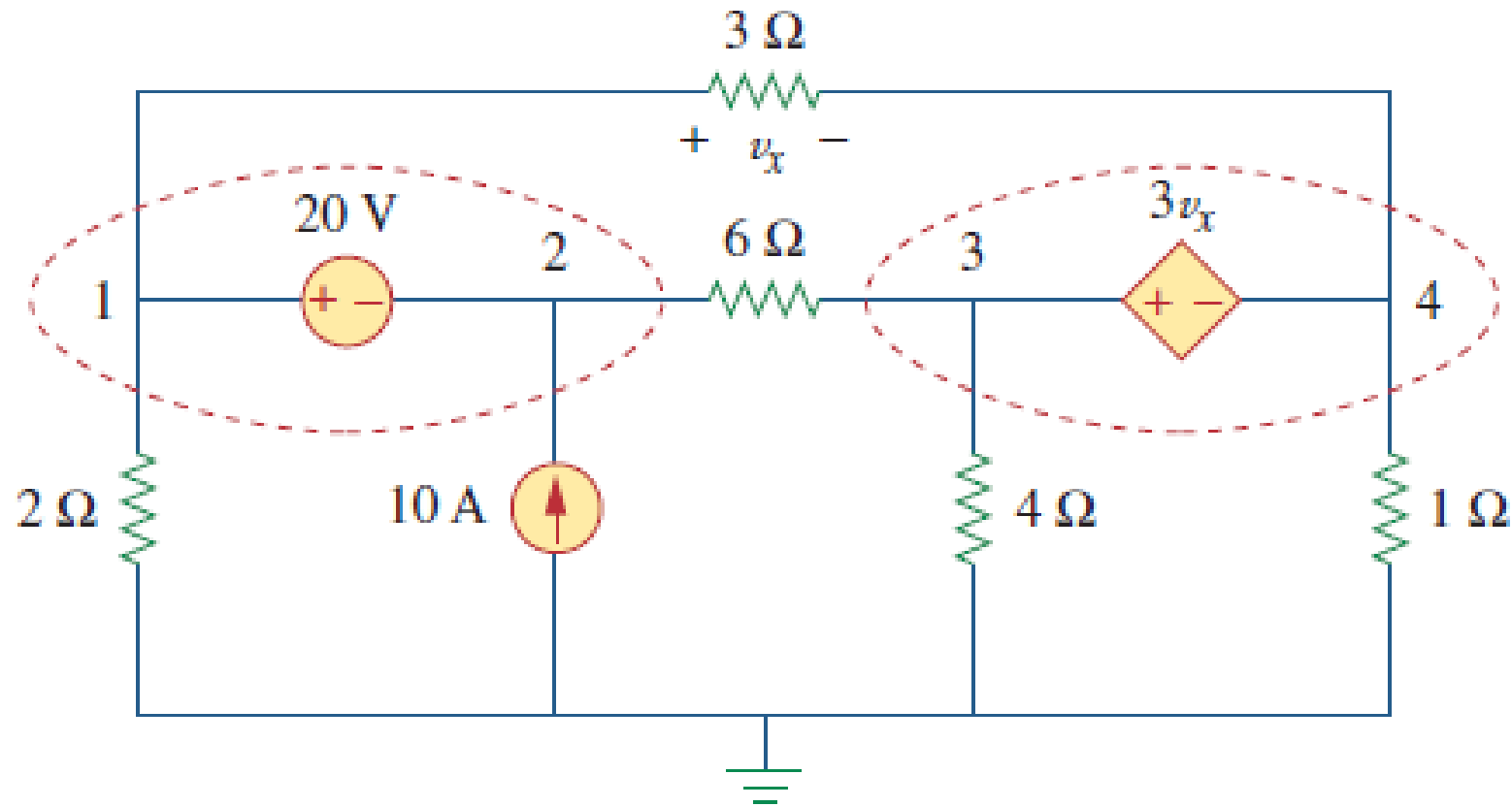
$$3v_1 = -22 \quad \Rightarrow \quad v_1 = -7.333 \text{ V}$$

$$\text{and } v_2 = v_1 + 2 = -5.333 \text{ V.}$$



Tutorial - 2

Determine the node voltages in the circuit.



Solution:

Nodes 1 and 2 form a supernode; so do nodes 3 and 4. We apply KCL

At supernode 1-2,

$$i_3 + 10 = i_1 + i_2$$

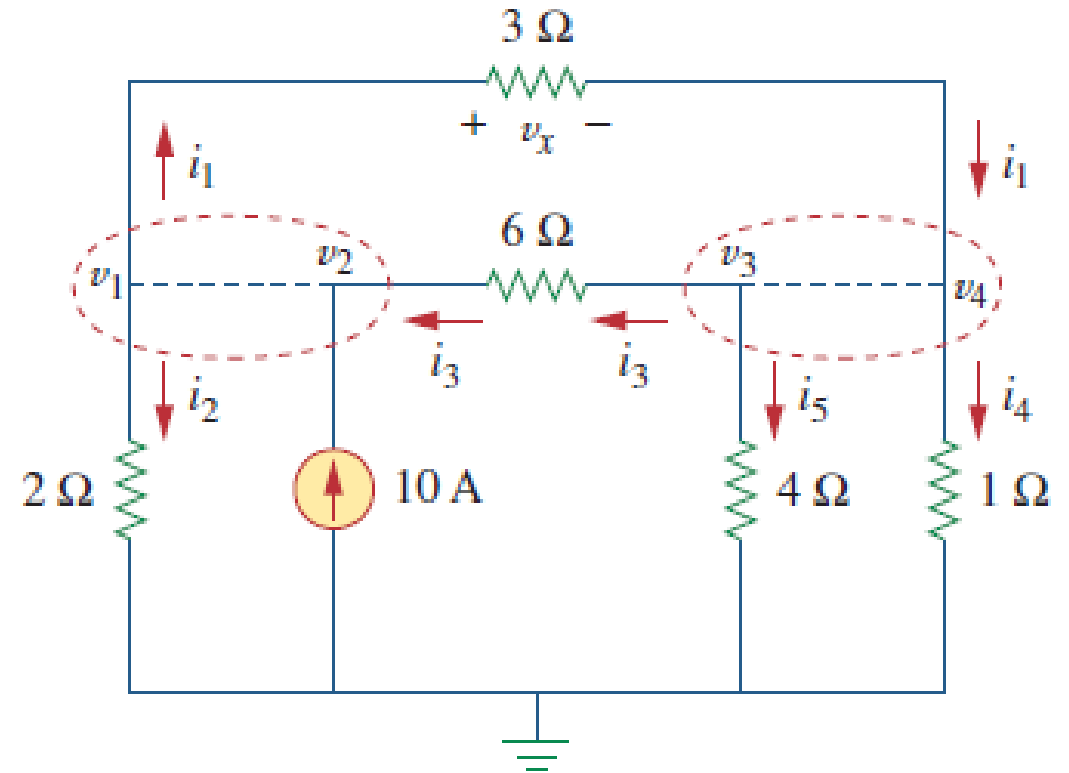
$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_4}{3} + \frac{v_1}{2}$$

$$5v_1 + v_2 - v_3 - 2v_4 = 60$$

At supernode 3-4,

$$i_1 = i_3 + i_4 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_4}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

$$4v_1 + 2v_2 - 5v_3 - 16v_4 = 0$$



For loop 1,

$$-v_1 + 20 + v_2 = 0$$

$$v_1 - v_2 = 20$$

For loop 2,

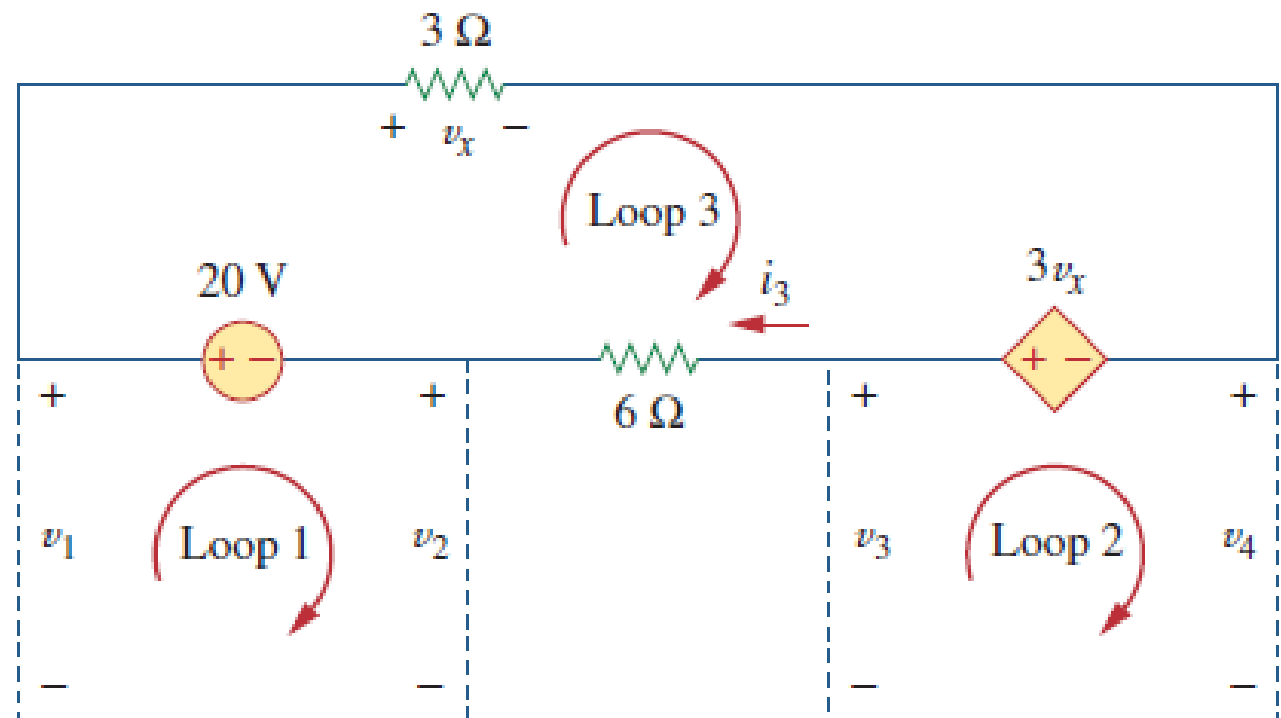
$$-v_3 + 3v_x + v_4 = 0$$

For loop 3,

$$v_x - 3v_x + 6i_3 - 20 = 0$$

But $6i_3 = v_3 - v_2$ and $v_x = v_1 - v_4$.

$$-2v_1 - v_2 + v_3 + 2v_4 = 20$$



$$v_1 = 26.67\text{ V}$$

$$v_2 = 6.667\text{ V}$$

$$v_3 = 173.33\text{ V}$$

$$v_4 = -46.67\text{ V}$$